

AIGR Deep Research — AprA (Q72DT2)

plasma membrane localization

(GO:0005886)

Gene: aprA — adenylyl-sulfate reductase, alpha subunit **Organism:** *Nitratidesulfovibrio* (*Desulfovibrio*) *vulgaris* Hildenborough (NCBITaxon:882) **UniProt:** Q72DT2 (664 aa; flavoprotein; SdhA/FrdA/AprA superfamily) **Focus:** function_assignment — does AprA directly localize to the plasma membrane (GO:0005886)?

Executive Judgment

Verdict: Partially supported / weakly supported as a bare CC term — refine, do not accept as-is.

AprA is fundamentally a **soluble cytoplasmic flavoenzyme**: the exact-strain enzyme was purified to homogeneity as a soluble $\alpha\beta$ ($\alpha_2\beta_2$) protein (

P 8174563

and Q72DT2 has **no transmembrane helix**. The plasma-membrane annotation currently carried by UniProt (GO:0005886) rests only on **IEA:TreeGrafter**, a phylogenetic graft from the SdhA/FrdA membrane-anchored flavoprotein family, and is therefore weak and imprecise.

However, the hypothesis is not simply wrong. There is genuine **experimental** evidence for a **peripheral, partially stable membrane-associated pool** of AprAB, recruited to the cytoplasmic face of the **integral membrane QmoABC complex** (co-purification + blue-native PAGE in the exact strain,

P 23842468

direct electron transfer,

P 26768116

Crucially, this is a **peripheral/extrinsic association driven by protein–protein interaction, not intrinsic membrane residency**. Membrane interaction via Qmo alone does not establish residency.

Bottom line for the curator: The primary location is **cytoplasm (GO:0005737)**. A membrane pool is real but peripheral; if captured, it is best modeled as **extrinsic component of the cytoplasmic side of the plasma membrane (GO:0031234)** supported by IPI evidence, not as bare "plasma membrane." Separately, the co-annotated **GO:0000104 succinate dehydrogenase activity** is a paralog overannotation and should be removed.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
P 8174563 (Verhagen 1994)	Direct biochemistry / purification	Refutes intrinsic membrane residency; supports soluble	Is AprAB a soluble enzyme?	Purified to homogeneity as soluble $\alpha\beta$ (67.8/25.6 kDa), $\alpha 2\beta 2$, one Fe-S cluster + FAD; no membrane anchor	<i>D. vulgaris</i> Hildenborough (exact strain)
P 23842468 (Krumholz 2013)	Interaction / co- purification / BN- PAGE	Qualifies (supports peripheral membrane pool)	Does AprAB associate with the membrane?	Membrane band contains 3 Qmo subunits + AprA + AprB; AprA/ QmoA/QmoB tagging reciprocally co- purifies AprA/B; "Qmo-Apr complex at least partially stable in protein extracts"	<i>D. vulgaris</i> Hildenborough + <i>D. alaskensis</i> G20
P 26768116 (Duarte 2016)	Direct assay (electrochemistry)	Qualifies (interaction, not residency)	Is the AprAB- Qmo link functional?	Direct QmoABC→AprAB electron transfer; Qmo essential for efficient electron delivery to sustain catalysis	<i>D. desulfuricans</i> ATCC 27774
	Mutant phenotype	Supports Qmo being the	Is Qmo the transmembrane	Δ qmoABC cannot grow on	<i>D. vulgaris</i> Hildenborough

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
P 20581180 (Zane 2010)		membrane partner	electron conduit to Apr?	sulfate; Qmo is a transmembrane complex delivering electrons to Apr	
UniProt Q72DT2 (database)	Computational / database	Competing (weak basis for GO:0005886; flags overannotation)	What is the current annotation basis?	GO:0005886 = IEA:TreeGrafter; GO:0000104 SDH activity = IEA:TreeGrafter; family = SdhA/FrdA/AprA (PF00890); 0 TM	Sequence/ family

GO Curation Implications (leads — require curator verification)

GO term	Aspect	Current basis	Recommended action	Rationale
GO:0005886 plasma membrane	CC	IEA:TreeGrafter	Do not retain as bare term / generalize-then-specify. Replace with GO:0031234 extrinsic component of cytoplasmic side of plasma membrane (or GO:0019898 extrinsic component of membrane) with IPI evidence from P 23842468 / 26768116.	AprAB is peripherally recruited via Qmo; no TM helix; bare "plasma membrane" overstates residency.
GO:0005737 cytoplasm	CC	(absent)	Add as primary location (IDA/IEA)	Soluble $\alpha\beta$ enzyme purified from non-membrane fraction (P 8174563); no membrane anchor.
GO:0000104 succinate dehydrogenase activity	MF	IEA:TreeGrafter	Remove	Paralog overannotation from SdhA/FrdA/AprA family; AprA catalyzes APS reduction, not succinate oxidation.
GO:0009973 adenylyl-sulfate reductase activity	MF	IEA:UniProtKB-EC	Retain (core function)	Correct primary molecular function.
GO:0009055 electron transfer activity	MF	IEA:TreeGrafter	Retain (defensible)	Consistent with FAD/Fe-S redox chemistry and Qmo electron transfer.

Avoids "protein binding" as the recommendation; the interaction is captured more informatively via the extrinsic-membrane CC term plus retention of the APS reductase MF term.

Computed Provenance (Iteration 2 sequence analysis)

Kyte–Doolittle hydropathy analysis of the Q72DT2 sequence (664 aa) fetched live from UniProt:

Metric	Value	Interpretation
Max KD hydropathy, 19-res window	1.232 (res 25–43)	Below the ~1.6 transmembrane threshold → no TM-length hydrophobic segment
N-terminal max window (res 1–30)	1.232	No signal-peptide-like hydrophobic n-region
GRAVY (grand avg hydropathy)	−0.371	Hydrophilic → soluble character
UniProt annotated TM features	0	Consistent with the above

Provenance CSV saved to `/tmp/aprA_artifacts/seq_membrane_analysis.csv`. This independent sequence check corroborates that AprA has **no intrinsic membrane anchor or secretion signal**; any plasma-membrane association is necessarily peripheral/extrinsic (Qmo-mediated), not integral.

QuickGO annotation provenance (live query, EBI): All 7 GO annotations for Q72DT2 are **electronic (IEA)**; **none are experimental**. Critically, GO:0005886 plasma membrane carries the strong `located_in` (residency) qualifier but is assigned solely by **TreeGrafter/PANTHER (GO_REF:0000118)** — the same source as the erroneous GO:0000104 succinate dehydrogenase, GO:0050660 FAD binding, GO:0009055 electron transfer, and GO:0009061 anaerobic respiration. Only GO:0009973 (adenylyl-sulfate reductase, UniProt EC, GO_REF:0000003) and GO:0016491 (oxidoreductase, InterPro) come from non-PANTHER sources.

GO id	Aspect	Qualifier	Evidence	AssignedBy / Ref
GO:0005886 plasma membrane	CC	located_in	IEA	TreeGrafter / GO_REF:0000118
GO:0000104 succinate dehydrogenase activity	MF	enables	IEA	TreeGrafter / GO_REF:0000118
GO:0009055 electron transfer activity	MF	enables	IEA	TreeGrafter / GO_REF:0000118
GO:0050660 FAD binding	MF	enables	IEA	TreeGrafter / GO_REF:0000118
GO:0009061 anaerobic respiration	BP	involved_in	IEA	TreeGrafter / GO_REF:0000118
GO:0009973 adenylyl-sulfate reductase activity	MF	enables	IEA	UniProt / GO_REF:0000003
GO:0016491 oxidoreductase activity	MF	enables	IEA	InterPro / GO_REF:0000002

The key curation point: despite direct experimental interaction/complex data existing (

P 23842468

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P 26768116

the CC annotation is a pure phylogenetic graft asserting residency (`located_in`). It is both **under-evidenced and over-specified** — the experimental record supports a peripheral, Qmo-recruited pool (an IPI-supportable *extrinsic component* CC), not intrinsic plasma-membrane residency.

Mechanistic Scope

- **Immediate molecular function (direct):** two-electron reduction of adenosine-5'-phosphosulfate (APS) → sulfite + AMP, using FAD and a [4Fe-4S] cluster (AprA = flavoprotein subunit; AprB = Fe-S subunit).

- **Immediate cellular context (direct):** soluble $\alpha\beta$ enzyme in the cytoplasm that **transiently/peripherally docks onto the cytoplasmic face of the membrane-integral QmoABC complex** to receive electrons from the menaquinone pool.
 - **Downstream/inferred (not localization evidence):** dissimilatory sulfate reduction growth phenotype; Δ qmoABC lethality on sulfate. These reflect the pathway, not AprA membrane residency.
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Conflicts and Alternatives

1. **Peripheral pool vs. residency:** Membrane co-fractionation (Krumholz) genuinely conflicts with a purely cytoplasmic model, but is fully explained by peripheral recruitment to Qmo — not intrinsic membrane insertion. Both can be true (dual pool).
 2. **Paralog confusion (SdhA/FrdA):** The FAD_binding_2 family drives TreeGrafter to graft membrane-anchoring and succinate-dehydrogenase terms from SDH/FRD, where the flavoprotein subunit is tethered to membrane cytochromes. This is the likely origin of GO:0005886 and the certain origin of the erroneous GO:0000104.
 3. **Partition unknown:** No study quantifies the soluble:membrane ratio of AprAB in vivo, so "resident membrane protein" cannot be asserted.
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Knowledge Gaps

Gap	What was checked	Why it matters	Resolving experiment
Quantitative soluble vs. membrane partition of AprAB	Literature (P 8174563 soluble; 23842468 membrane pool) — no partition data	Determines whether "membrane" is a minor or major pool; affects whether a CC membrane term is warranted at all	Quantitative cell fractionation / Western of AprA in soluble vs. membrane fractions $\pm$ Qmo
Dependence of membrane pool on Qmo	Co-purification only	Confirms peripheral (Qmo-dependent) vs. independent membrane binding	AprA localization in Δ qmoABC background
Structural interface / topology of Apr–Qmo	Electrochemistry (26768116), no cryo-EM cited here	Confirms extrinsic cytoplasmic-face geometry	Cryo-EM of the Qmo–Apr supercomplex

Discriminating Tests

1. **Fractionation $\pm$ high salt / carbonate wash:** peripheral proteins strip off at high pH/salt — would confirm extrinsic (not integral) membrane association.
2. **AprA localization in Δ qmoABC:** loss of the membrane pool would prove Qmo-dependent peripheral recruitment.
3. **Cryo-EM / crosslinking of the Apr–Qmo supercomplex:** defines the cytoplasmic-face docking geometry.
4. **Family-aware GO re-annotation:** verify that GO:0005886/GO:0000104 trace to SDH/FRD ancestral grafts and are not experimentally supported for AprA.

Curation Leads (require curator verification)

- **Candidate references to attach:**

P 8174563

(soluble $\alpha\beta$ enzyme, exact strain),

P 23842468

(Qmo-Apr membrane co-complex, exact strain),

P 26768116

(direct Qmo→Apr electron transfer),

P 20581180

(Qmo membrane partner, phenotype).

- Verify snippet — 23842468: *"This provides clear evidence for the presence of a Qmo-Apr complex that is at least partially stable in protein extracts of D. vulgaris and D. alaskensis."*
 - Verify snippet — 8174563: *"Adenosine phosphosulfate reductase from Desulfovibrio vulgaris Hildenborough has been purified to homogeneity and was found to consist of two subunits."*
 - Verify snippet — 26768116: *"the QmoABC complex is essential for efficient electron delivery to AprAB, in order to sustain catalysis."*
 - **Candidate GO changes:** add GO:0005737 (cytoplasm, primary); replace bare GO:0005886 with GO:0031234 (extrinsic component of cytoplasmic side of plasma membrane, IPI); remove GO:0000104 (succinate dehydrogenase activity, paralog overannotation); retain GO:0009973.
 - **Suggested curator question:** Is there any quantitative fractionation establishing AprA as a majority membrane pool? If not, treat plasma-membrane residency as non-core / peripheral only.
 - **Suggested experiments:** carbonate-wash fractionation; AprA localization in Δ qmoABC.
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Limitations

Literature search was constrained to PubMed abstracts; UniProt was queried live (GO evidence codes, family, TM count). No local bioinformatics files were used. Snippets are quoted verbatim from retrieved abstracts and should be re-verified against source.

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